

Report Building Guide

Coding-Forge

Prerequisites

Before building reports, make sure you have:

- All 3 data files loaded and cleaned (see POWER_QUERY_TRANSFORMATIONS.md)
 - Relationships set up in Model View: Programs → Budget vs Actuals, Programs → Purchase Orders (see INSTRUCTIONS.md)
 - Power BI Desktop
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Report 1: Program Financial Health Dashboard

Audience: Program Managers, Finance Directors Purpose: Track budget vs. actual spend across all programs at a glance

Page Setup

1. Go to Report View (left sidebar)
2. Right-click the page tab at the bottom > Rename > **Program Health**

Visuals to Build

Visual 1: Clustered Bar Chart — Budget vs Actual by Program

1. Click on empty space on the canvas
2. From the Visualizations pane, click Clustered Bar Chart
3. Drag fields from the Fields pane:
 - Y-axis: **Budget vs Actuals** > **Program Name**
 - X-axis: **Budget vs Actuals** > **Budgeted Amount**
 - Also drag **Actual Amount** to X-axis (it will appear as a second bar)
4. Resize the visual to fill the top half of the page

Formatting:

- Click the visual > Format pane (paint roller icon)
- Turn on Data labels (shows the numbers on the bars)
- Under Title > change text to **Budget vs Actual by Program**

Visual 2: Card — Total Budget

1. Click empty canvas space
2. Click Card visual
3. Drag **Budgeted Amount** to the Fields area
4. It will automatically show the SUM
5. In Format pane > Callout value > Set display units to **Millions**

Visual 3: Card — Total Actual Spend

1. Repeat same steps with **Actual Amount**

Visual 4: Table — Detailed Breakdown

1. Click Table visual
2. Drag these fields in order:
 - **Program Name**
 - **Budget Category**
 - **Budgeted Amount**
 - **Actual Amount**
 - **Variance**
 - **Status**
3. Resize to fill the bottom half

Formatting the Table:

- Format pane > Style presets > Choose **Alternating rows** (this is now the default in April 2026+, but verify it's applied)
- Under Conditional formatting, right-click the **Status** column header in the visual > Conditional formatting > Background color:
 - Select Rules
 - If value contains **Over Budget** → Red
 - If value contains **At Risk** → Yellow
 - If value contains **On Track** → Green

Visual 5: Slicer — Filter by Fiscal Year

1. Click the Button Slicer visual in the Visualizations pane
2. Drag **Fiscal Year** field to the Field area
3. Place it in the top-right corner
4. Users can click FY2025 or FY2026 to filter the entire page

Visual 6: Slicer — Filter by Quarter

1. Click another Button Slicer visual
2. Drag **Quarter** field to the Field area
3. Place it next to the Fiscal Year slicer
4. Users can click a quarter to filter the entire page

Report 2: Procurement & Purchase Orders

Audience: Procurement Officers, Contracting Purpose: Monitor PO status, vendor spend, and delivery timelines

Page Setup

- Add a new page: Click the + at the bottom

- Rename to **Procurement**

Visuals to Build

Visual 1: Stacked Bar Chart — PO Amount by Vendor

1. Click Stacked Bar Chart
2. Y-axis: **Purchase Orders** > **Vendor Name**
3. X-axis: **Purchase Orders** > **PO Amount**
4. Legend: **Purchase Orders** > **PO Status**

This shows which vendors have the most spend and what status their POs are in.

Visual 2: Donut Chart — PO Status Distribution

1. Click Donut Chart
2. Legend: **PO Status**
3. Values: **PO ID** (it will count them automatically)
4. Title: **PO Status Breakdown**

Visual 3: Table — Outstanding POs

1. Click Table visual
2. Add fields: **PO ID**, **Vendor Name**, **PO Amount**, **Amount Remaining**, **Delivery Date**, **Priority**
3. Click the column header **Amount Remaining** in the table > Sort descending (shows largest outstanding amounts first)

Conditional Formatting on Priority:

- Right-click **Priority** column header > Conditional formatting > Background color > Rules:
- **Critical** → Red
- **High** → Orange
- **Medium** → Yellow
- **Low** → Green

Visual 4: Clustered Column Chart — Spend by Program

1. Click Clustered Column Chart
2. X-axis: **Program Name**
3. Y-axis: **PO Amount**
4. Title: **Procurement Spend by Program**

Visual 5: Slicer — Filter by Priority

1. Click the Button Slicer visual in the Visualizations pane (displays items as clickable tiles)
2. Drag **Priority** to the Field area
3. This creates a row of clickable buttons — users can select one or multiple priorities

Report 3: Executive Summary (Multi-Page Drillthrough)

Audience: Executives, Senior Leadership Purpose: High-level KPIs with ability to drill into details

Page Setup

- Add new page > Rename to **Executive Summary**

Visuals to Build

Row 1: KPI Cards (across the top)

Create 4 Card visuals side by side:

Card	Table	Field	Display Units
Total Budget	Budget vs Actuals	Budgeted Amount	Millions
Total Actual	Budget vs Actuals	Actual Amount	Millions
Total PO Value	Purchase Orders	PO Amount	Millions
Total Outstanding	Purchase Orders	Amount Remaining	Millions

Row 2: Program Status Overview

Visual: Matrix

1. Click Matrix visual
2. Rows: **Programs** > **Program Name**
3. Columns: **Budget vs Actuals** > **Quarter**
4. Values: **Budget vs Actuals** > **Actual Amount**
5. This gives a pivot-table style view of spending by program and quarter

Row 3: Trend Chart

Visual: Line Chart

1. Click Line Chart
2. X-axis: **Budget vs Actuals** > **Quarter**
3. Y-axis: **Budget vs Actuals** > **Actual Amount**
4. Legend: **Programs** > **Program Name**
5. Title: **Spending Trend by Program**

Setting Up Drillthrough

Drillthrough lets users right-click a program name on the Executive Summary page and jump to a detail page:

1. Add a new page > Rename to **Program Detail**
 2. In the Visualizations pane on that page, find the Drillthrough section at the bottom
 3. Drag **Program Name** into the Drillthrough field
 4. Now build detail visuals on this page (table of POs, budget breakdown, variance detail for that specific program)
 5. A Back button appears automatically — users click it to return to the Executive Summary
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Filtering Within Visuals

Power BI offers several ways to filter data directly within visuals, giving report consumers interactive control without needing separate slicers.

Visual-Level Filters (Filters Pane)

Each visual has its own filter area in the Filters pane on the right:

1. Select a visual on the canvas
2. In the Filters pane, you'll see three sections:
 - Filters on this visual — affects only the selected visual
 - Filters on this page — affects all visuals on the current page
 - Filters on all pages — affects the entire report
3. Drag any field into one of these sections
4. Set the filter type: Basic (checkboxes), Advanced (contains, starts with, etc.), or Top N

Example: Add **Budget vs Actuals** > **Status** to "Filters on this visual" on a table, then check only "Over Budget" and "At Risk" to show only problem areas.

Show/Hide the Filters Pane for Report Consumers

1. Select the Filters pane header
2. In the Format pane, toggle Filters pane > Allow end users to filter (on/off per filter)
3. You can also lock filters so users see them but cannot change them

Enable Visual Header Filter Icon

This adds a small funnel icon to the top-right of each visual that users can click to filter:

1. Go to File > Options and settings > Options
2. Under Current file > Report settings
3. Ensure Persistent filters is enabled
4. For individual visuals: Click a visual > Format pane > General > Header icons > Turn on Filter icon

Cross-Filtering Between Visuals (Visual Interactions)

Clicking a data point in one visual automatically filters the others on the same page:

1. Click on a visual (e.g., the Clustered Bar Chart)
2. Go to Format tab in the ribbon > Edit interactions
3. Small icons appear on every other visual:
 - Filter (funnel) — the other visual filters to show only related data
 - Highlight — the other visual grays out non-related data but still shows totals
 - None — no interaction
4. Click the desired icon on each visual to control how they respond

Recommended interactions for this report:

- Bar Chart clicking → Filters the Table (shows only that program's details)
- Donut Chart clicking → Highlights the Bar Chart (shows proportion without losing totals)
- KPI Cards → Set to None (cards should always show overall totals)

Adding a Visual-Level Filter Example (Step-by-Step)

To add a "Top 3 over-budget programs" filter on the Table visual:

1. Click the Table visual
 2. In Filters on this visual, drag **Variance** field
 3. Change filter type to Top N
 4. Select Bottom 3 (lowest variance = most over budget)
 5. Drag **Variance** to the "By value" area
 6. Click Apply filter
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General Formatting Tips

Apply a Consistent Theme

1. View tab > Themes > Browse for a theme or select a built-in one
2. For Coding-Forge branding, consider creating a custom theme with company colors

Add Page Navigation Buttons

1. Insert tab > Buttons > Navigation > Page navigator
2. This adds buttons at the top that let users click between pages

Add a Title Banner

1. Insert tab > Text box
2. Type your report title (e.g., "Coding-Forge Program Financial Dashboard")
3. Format: Bold, larger font, company blue background

Refresh Schedule (After Publishing)

In Power BI Service:

1. Go to the dataset in your workspace
 2. Settings > Scheduled refresh
 3. Set to refresh daily or as needed (depends on Dataverse gateway configuration)
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For data cleanup steps, see `POWER_QUERY_TRANSFORMATIONS.md`. For connection and setup details, see `INSTRUCTIONS.md`.